

ZUWARA, Libyan Arab JamahiriyaWMO: **620070**Lat: **32.883N**Lon: **12.083E**Elev: **13**StdP: **14.69**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.1	48.1	31.1	25.3	56.9	33.8	28.4	57.3	28.0	60.3	24.9	60.1	8.8	240	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	96.4	73.5	91.7	75.7	88.7	77.3	82.8	88.2	81.4	86.6	80.2	85.6	9.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.5	86.9	79.9	155.1	85.4	78.7	149.1	84.5	46.7	88.3	45.1	86.6	43.8	85.8	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	21.9	19.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	58.7	59.0	62.3	66.0	71.8	76.9	81.3	83.4	80.8	74.9	66.1	59.7
	DBStd	9.99	3.37	4.64	5.72	5.34	5.36	4.45	3.18	2.96	3.58	5.06	5.35	5.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	776	199	180	122	38	3	0	0	0	0	1	50	183
	CDD50	7349	269	252	383	481	675	806	969	1035	925	770	482	302
	CDD65	2650	3	12	40	70	213	356	504	570	475	306	81	20
	CDH74	23868	24	94	316	455	960	2328	5286	6837	4963	2139	350	116
	CDH80	7756	2	23	147	226	411	559	1608	2645	1514	529	67	25
(14) Wind	WSAvg	9.8	8.4	9.4	9.9	11.2	10.9	10.8	9.6	10.2	10.7	9.1	8.5	8.9
(15) Precipitation	PrecAvg	9.4	1.4	0.7	0.7	0.5	0.2	0.0	0.0	0.0	0.8	1.7	1.5	1.8
	PrecMax	17.9	6.4	2.5	3.1	3.3	2.9	0.2	0.0	0.4	5.1	6.5	7.5	6.8
	PrecMin	1.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.9	1.2	0.7	0.8	0.7	0.5	0.0	0.0	0.1	1.2	1.7	1.8	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	83.4	92.2	96.7	103.1	99.7	98.6	101.1	101.8	95.9	85.6	83.2
		MCWB	58.3	64.2	69.6	68.2	77.1	71.9	76.9	75.6	72.7	71.7	66.3	62.8
	2%	DB	71.8	76.0	82.0	85.4	90.6	89.6	90.8	92.7	91.8	89.0	81.2	76.8
		MCWB	57.5	59.8	63.3	66.3	70.1	74.3	77.8	79.4	76.8	73.3	64.4	62.0
	5%	DB	68.0	70.5	74.1	77.3	81.7	85.0	87.9	89.6	88.0	84.6	77.7	71.4
		MCWB	55.6	58.2	61.0	64.1	69.0	74.1	78.0	80.0	77.8	74.9	66.7	60.1
	10%	DB	65.4	66.6	69.7	72.6	77.4	82.1	86.0	87.8	86.0	82.0	74.9	68.2
		MCWB	56.7	57.1	60.9	63.4	68.7	73.5	77.4	79.3	77.1	73.9	66.1	59.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	68.0	70.7	73.8	82.0	80.4	83.5	86.7	83.1	81.7	73.8	68.3
		MCDB	68.6	81.1	81.8	84.7	97.0	89.1	89.7	93.1	88.2	86.4	78.3	74.4
	2%	WB	61.2	63.0	66.0	68.8	73.6	78.0	80.8	83.6	81.5	79.3	71.3	64.7
		MCDB	66.5	70.4	75.2	79.6	83.7	84.7	86.3	88.3	86.5	83.5	76.0	71.1
	5%	WB	59.5	60.4	63.3	66.2	71.1	76.3	79.7	81.9	80.1	77.4	69.2	62.8
		MCDB	64.7	66.7	70.0	74.4	77.6	82.0	85.3	86.7	85.4	81.8	74.9	69.2
	10%	WB	58.2	58.8	62.0	64.7	69.7	75.0	78.7	80.9	79.0	75.4	67.2	60.7
		MCDB	63.5	64.3	67.7	70.9	75.7	80.2	84.4	86.0	84.4	80.7	73.7	66.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	14.2	13.7	11.9	10.7	9.5	9.2	9.3	10.0	12.6	15.0	14.9
		MCDBR	20.1	22.3	24.3	23.2	21.1	16.2	13.3	13.1	14.6	17.3	18.6	20.2
		MCWBR	12.5	13.8	12.2	11.3	11.2	9.3	9.0	9.4	9.7	11.6	11.7	13.1
	5% WB	MCDBR	14.3	17.8	17.0	17.3	14.1	12.6	10.7	11.2	10.7	13.2	14.6	16.5
		MCWBR	11.6	13.4	11.4	10.4	9.7	9.0	8.7	8.7	8.8	10.8	12.1	12.9
(40) Clear-Sky Solar Irradiance	taub		0.389	0.414	0.495	0.551	0.553	0.501	0.507	0.503	0.528	0.484	0.435	0.398
	taud		2.192	2.108	1.893	1.789	1.815	1.964	1.967	1.990	1.931	2.003	2.079	2.158
	Ebn at Noon		255	261	248	240	241	253	251	250	236	235	235	243
	Edh at Noon		37	45	60	69	68	58	58	56	57	49	41	36
(44) All-Sky Solar Radiation	RadAvg		988	1265	1637	1977	2199	2396	2423	2182	1773	1364	1048	868
	RadStd		49	62	79	103	102	84	55	45	68	76	53	62

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-95	+209	N/A

Nomenclature: See separate page